

GRADE 9 CHEMISTRY SYLLABUS FOCUS

Interrelated Topics: The Mole Concept, Redox Reactions & Quantitative Stoichiometry

CURRICULUM OVERVIEW & TOPIC INTERRELATION

Academic Level: Grade 9 Chemistry Honors / Pre-Advanced Syllabus

Faculty Lead: Dr. Eleanor Vance | St. Jude Preparatory Academy

Core Theme: Quantitative Electron Transfer & Mass Conservation

This syllabus deliberately links three foundational pillars of modern chemistry:

- 1) The Mole Concept (counting discrete particles by weighing bulk mass)
- 2) Redox Reactions (tracking fundamental electron exchange across oxidation states)
- 3) Stoichiometric Redox Analysis (quantifying exact electron and mass balance)

TOPIC 1: THE MOLE CONCEPT & MOLAR MASS BRIDGES

Target Mastery: 100% Quantitative Fluency

- Avogadro's Constant: $N_A = 6.022 \times 10^{23}$ particles per mole
 - Molar Mass Calculation (M): Summation of atomic weights from Periodic Table (g/mol)
 - Fundamental Conversion Formula: $n = \text{mass (g)} / \text{Molar Mass (g/mol)}$
 - Particle Counting Bridge: $\text{Number of Molecules} = n \times 6.022 \times 10^{23}$
 - Molar Gas Volume at STP: 1 mole of any ideal gas occupies 22.4 dm^3 (Liters)
- Key Diagnostic Trap: Applying coefficient multipliers directly to gram masses instead of moles

TOPIC 2: OXIDATION-REDUCTION (REDOX) REACTIONS

Target Mastery: Electron Transfer & State Identification

- Fundamental Concept: Oxidation Is Loss of electrons, Reduction Is Gain (OIL RIG)
 - Oxidation Number Assignment Rules: Free elements = 0, Group 1 = +1, Oxygen = -2, Hydrogen = +1
 - Identifying Oxidizing Agents (electron acceptors) and Reducing Agents (electron donors)
 - Construction of Half-Equations with explicit electron representation (e^-)
 - Disproportionation & Synthesis Redox Transformations in aqueous solution
- Key Diagnostic Trap: Confusing oxidation state increases with electron gain (charge inversion)

TOPIC 3: STOICHIOMETRY & QUANTITATIVE REDOX ANALYSIS

Target Mastery: Mass, Electron & Yield Conservation

- Electron Balance Principle: Total electrons lost in oxidation MUST equal electrons gained in reduction
 - Balancing complex Redox Equations using electron-conservation half-reaction methods
 - Mole-to-Mole Stoichiometric Ratios derived from balanced redox equations
 - Limiting Reagent Determination: Using mole-per-coefficient ratios to isolate the limiting species
 - Excess Reactant Accounting: $\text{Moles remaining} = \text{Initial moles} - \text{Consumed moles}$
 - Percent Yield Calculations: $\text{Percent Yield} = (\text{Actual Mass} / \text{Theoretical Mass}) \times 100\%$
- Key Diagnostic Trap: Failure to subtract consumed moles from initial reactant pools